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The Editors
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Dear Editors,

We submit for your consideration as a Paper “Calibration-derived decoder discriminability is associated with online P300-speller accuracy, but the fitted mapping does not transport across cohorts.” A calibration-derived score has repeatedly been related to online P300-speller accuracy within the cohort where it was measured, including by Mainsah and colleagues in this journal (*J Neural Eng* 2016;13:066007), who derived speller accuracy analytically from a calibration-derived detectability index. This manuscript asks whether such a mapping, once fitted, transports to a cohort it has never seen.

Using the public BigP3BCI archive, we evaluated a calibration-derived score across 18 source-study cohorts with a leave-one-study-out design (271 participants, 739 session-condition records, 19,611 selections), rather than within a single cohort, as every prior report has done. Each withheld cohort’s expected accuracy was estimated purely from the other 17.

The association was positive in every withheld cohort, though precision varied enough that six of the eighteen 95% confidence intervals included zero; pooled at the participant level the relationship was precisely estimated ($r = 0.714$, $p < 0.001$). The fitted mapping did not transport: the calibration intercept had a between-cohort standard deviation of tau = 0.87 (95% interval for an unrepresented cohort, -1.97 to 1.85 log-odds), and the slope showed the same pattern (tau = 0.43, interval 0.111 to 2.005). Both conclusions hold at the lower confidence limit of tau and under an independent bootstrap estimate. A four-cohort ALS-only subgroup gave a materially tighter, more favourable picture (uncorrected slope spread 0.223, against 0.560 across all 18), demonstrating how few-cohort evaluation understates real-world variability.

The manuscript separates two claims often reported together: an association present in every cohort, and a mapping that transports between them. The score may support ranking sessions within a setting after local validation; it should not be used to report expected accuracy in a cohort where the mapping was not developed, without local recalibration.

This manuscript is original, is not under consideration elsewhere, and has not been posted as a preprint. The authors declare no competing interests.

Sincerely,
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